

- A. What is the difference between EREW, CREW, and CRCW models? Provide an example for each of them. (1)

- B. How can the prefix sum of an array be computed on a CREW PRAM using idle processors from the sum algorithm? (1)

- C. Given an algorithm with $T_1(n)=n^2$ and $T_p(n)=n^2/p$, analyze the speedup and efficiency. What can you conclude? (1)

- D. What is the main conceptual difference between Amdahl's Law and Gustafson's Law? (1)

- A. Explain the difference between superscalar execution and SIMD execution. (1)

- B. Why does multi-threading help hide memory latency, and how many threads are needed to achieve full utilization if each thread performs 3 arithmetic operations with a latency of one cycle, followed by a memory load with 12-cycle latency? Please motivate the answer. (1)

- C. What does it mean for a computation to be bandwidth-bound, and why is vector multiplication of large arrays an example of this? (1)

- D. Write a short pseudocode that would result in worst-case SIMD efficiency on an 8-wide SIMD processor due to divergent execution. (1)

- A. Please describe shared address space model in terms of communication abstraction and required hardware support. (2)

- B. Why must CUDA allocate execution contexts for all threads in a block before execution begins? (2)

- A. What is the difference between joining and waiting on a barrier in Pthreads? (1)

- B. Give two examples of OpenMP environment variables and how they influence program execution. (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

- C. Describe how each thread in the following diagram behaves at runtime as it encounters points marked with 1, representing `pragma omp cancel`, and with 2, representing `pragma omp cancellation point`. (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**


